SOFTWARE REQUIREMENTS SPECIFICATION

Simplified Billing & Inventory Management System

Offline Desktop Application

Functional Requirements, Non-Functional Requirements and Solution Architecture

DOCUMENT VERSION
1.0 — Proposed Baseline

PREPARED ON
31 July 2026

PRIMARY DEPLOYMENT
Single-shop, single-workstation, offline

TARGET USERS
Small retail and grocery shops

TECHNOLOGY BASELINE
Java 21, Spring Boot, React, MySQL

DOCUMENT STATUS
For review and approval

DOCUMENT CONTROL

Purpose and Contents

This document defines the proposed functional scope, quality attributes, business rules, deployment model and acceptance targets for a desktop-installable billing and inventory application that performs all core operations without internet access.

> **Baseline decision:** The first release is one authoritative application, one local Spring Boot service and one local MySQL database on a single shop computer. Optional multi-terminal LAN operation and cloud synchronization are outside the MVP.

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Requirement notation

Notation	Meaning
FR-xxx	Functional requirement defining system behavior.
BR-xxx	Mandatory business rule or data-integrity constraint.
NFR-xxx	Measurable non-functional or quality requirement.
Must	Required for MVP acceptance.
Should	Important but may be scheduled after the initial MVP.
Could	Optional enhancement.

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Executive Summary

The product is a fast, keyboard-first billing and stock-control application for small shops. It replaces handwritten bills and disconnected spreadsheets with a reliable local transaction system while keeping operation simple for cashiers and owners.

Primary outcomes

- ✓ Complete sales rapidly using a barcode scanner.
- ✓ Maintain accurate stock automatically.
- ✓ Track purchases, returns and stock movements.
- ✓ Track customer Udhaar through an immutable Khata ledger.
- ✓ Print 58 mm and 80 mm thermal receipts.
- ✓ Operate without internet or a cloud account.
- ✓ Recover safely from crashes, printer failures and updates.

Product principles

- ✓ Scanner and keyboard first.
- ✓ Server-authoritative financial calculations.
- ✓ Atomic checkout and stock deduction.
- ✓ No deletion of completed financial history.
- ✓ Local ownership of business data.
- ✓ Simple MVP with explicit future boundaries.
- ✓ Backups treated as a core workflow.

Success measures

Measure	Target outcome
Normal barcode sale	Cashier completes the workflow without using a mouse.
Internet dependency	Zero internet dependency for installation and core daily operation.
Checkout integrity	No partial invoice, payment, stock or Khata updates.
Duplicate prevention	Repeated checkout submission creates no duplicate sale.
Disaster readiness	Owner can create, verify and restore an encrypted external backup.
Operational clarity	UI clearly distinguishes sale failure, sale success and print failure.

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Scope and Assumptions

2.1 MVP scope

Included

- Offline desktop installation and setup
- Local users, roles and authentication
- Shop and receipt configuration
- Products, categories, barcodes and stock ledger
- Suppliers, purchases and purchase returns
- POS, payments, GST, discounts and billing
- Sales cancellation and controlled sales returns
- Customers, Udhaar and Khata settlements
- Dashboard, operational reports and exports
- Thermal printing and reprinting
- Encrypted backup, restore, updates and diagnostics

Deferred

- Multi-store operation
- Multiple independent offline databases and synchronization
- Cloud backup or cloud reporting
- Online payment-gateway confirmation
- Automatic WhatsApp or SMS delivery
- E-invoice and e-way bill integrations
- Loyalty schemes and promotional engines
- Native mobile applications
- Direct integrations with all weighing-scale models

2.2 Operating assumptions

- The first release operates on one shop workstation with one local database.
- Barcode scanners normally emulate keyboard input and terminate scans with Enter.
- Thermal printers are installed through the operating-system print subsystem.
- UPI and card payments are recorded after the cashier confirms payment externally; the MVP does not verify them online.
- The owner is responsible for connecting an external drive or selecting a separate backup destination.
- The application stores money in INR by default but keeps currency configurable.
- GST behavior is configurable because some shops or products may be non-GST.

2.3 Constraints

- Backend: Java 21 and Spring Boot 3.x modular monolith.
- Persistence: MySQL 8.4 LTS with Spring Data JPA and Flyway migrations.
- Desktop UI: React with Tailwind CSS hosted in an Electron shell.
- Security: local authentication, role-based access and protected loopback API.
- Core assets, libraries, fonts and documentation must be packaged locally.
- Business-critical records are immutable and corrected through reversal transactions.

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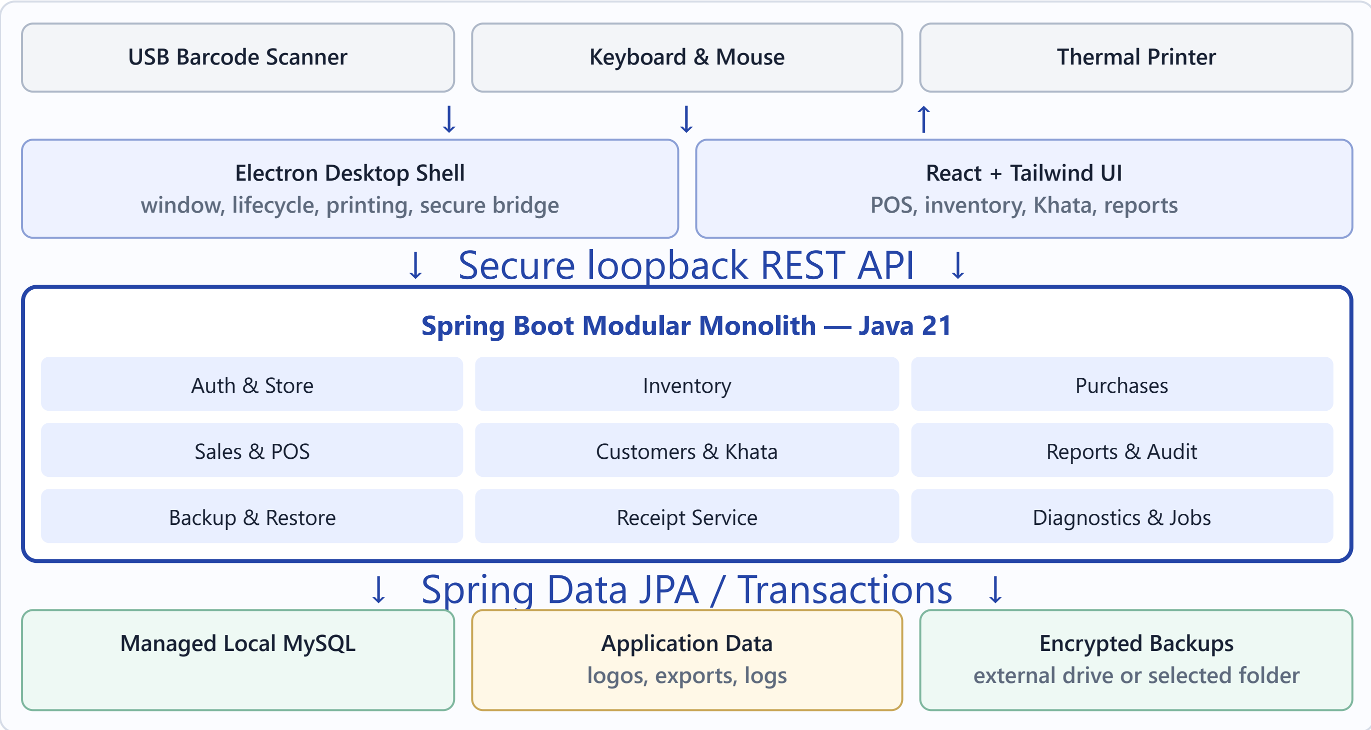
Technology and Solution Architecture

3.1 Runtime architecture



3.2 Architectural style

The backend is a modular monolith organized by business feature. Each module contains its controller, DTO, application service, domain model and repository. A module may call another module only through an exposed service interface; it must not access another module's repositories directly.

Layer responsibilities

- **Controller:** HTTP mapping, authentication context and DTO validation.
- **Application service:** Use-case orchestration and transaction boundaries.
- **Domain:** Business rules, calculations, states and invariants.
- **Repository:** Persistence and locked queries.
- **Infrastructure:** printing, backups, files and OS integration.

Cross-cutting components

- JWT/session security and role checks
- Bean Validation and consistent API errors
- Audit and correlation identifiers
- Central money, tax and rounding policies
- Flyway schema migrations
- Health checks, rotating logs and diagnostics

3.3 Checkout consistency boundary

1. Submit
Cart + payment + idempotency key

›

2. Validate
User, products, price, tax and stock

›

3. Commit
Invoice, payment, stock and Khata atomically

›

4. Print
Only after database commit

> A printer failure never cancels a committed sale. The invoice remains searchable and reprintable. A database failure rolls back the complete checkout.

3.4 Deployment profile

Component	Deployment	Network exposure
Electron + React	Installed desktop application	No remote navigation required
Spring Boot	Application-managed local process/service	Loopback interface only
MySQL	Application-managed local instance	Loopback interface only
Backups	Selected local, external or network folder	No cloud dependency
Updates	Signed offline installation package	No forced online update

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Roles and Permissions

Capability	Owner/Admin	Cashier	Inventory Manager	Viewer
Use POS and print receipts	Yes	Yes	Optional	No
Apply discount within configured limit	Yes	Yes	No	No
Override price or discount limit	Yes	No	No	No
Manage products and purchases	Yes	No	Yes	No
Post stock adjustments	Yes	No	Yes	No
Cancel invoice or approve return	Yes	No	No	No
Record Khata settlement	Yes	Yes	No	No
View purchase cost and profit	Yes	No	Yes	Yes
Configure users, backup and restore	Yes	No	No	No
View reports	Yes	Limited	Inventory	Yes

> Permissions must be enforced by Spring Security on the backend. Hiding a button in the desktop UI is not an authorization control.

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Functional Requirements

5.1 Installation and application lifecycle

ID	Requirement	Priority
FR-INS-01	Provide a guided offline installer that includes all required application runtimes and initializes the local database.	**Must**
FR-INS-02	Installation shall not require separately installed Java, Node.js, MySQL, Docker or internet access.	**Must**
FR-INS-03	Create desktop and Start-menu shortcuts and record application, database and schema versions.	**Must**
FR-INS-04	On launch, start or connect to MySQL, start Spring Boot, verify health, run applicable migrations and then open the login screen.	**Must**
FR-INS-05	Prevent conflicting application instances from managing the same local database.	**Must**
FR-INS-06	Provide a recovery screen for startup failures with retry, restore, log export and safe shutdown actions.	**Must**
FR-INS-07	Preserve business data by default during uninstall and require separate confirmation before data removal.	**Must**

5.2 Store setup, users and authentication

ID	Requirement	Priority
FR-AUT-01	A first-run wizard shall capture shop identity, GST settings, timezone, invoice format, receipt width, printer, backup destination and first administrator.	**Must**
FR-AUT-02	Authenticate users entirely against local records and support administrator, cashier, inventory manager and viewer roles.	**Must**
FR-AUT-03	Support secure password changes, administrator-controlled password reset and a configured offline owner-recovery method.	**Must**
FR-AUT-04	Lock or throttle accounts after repeated failed sign-in attempts and audit authentication events.	**Must**
FR-AUT-05	Lock the application after configurable inactivity and support rapid cashier switching using a password or PIN.	**Should**
FR-AUT-06	Require administrator approval for price overrides, excessive discounts, cancellations, returns, restores and sensitive configuration changes.	**Must**

5.3 Product and inventory management

ID	Requirement	Priority
FR-INV-01	Create, view, edit, deactivate, search, sort and page products and categories.	**Must**
FR-INV-02	Maintain product name, receipt name, barcode, SKU, category, unit, HSN, GST rate, purchase cost, selling price, stock and minimum-stock level.	**Must**
FR-INV-03	Support piece, kilogram, gram, litre, millilitre, packet, box and dozen units, including decimal quantities for loose products.	**Must**
FR-INV-04	Generate unique internal barcodes without a remote service and print configurable barcode labels.	**Should**
FR-INV-05	Record every stock change as an immutable ledger transaction with type, quantity, reference, user, timestamp and reason.	**Must**

ID	Requirement	Priority
FR-INV-06	Display low-stock and out-of-stock alerts and calculate reorder suggestions.	**Must**
FR-INV-07	Support opening-stock import, CSV import preview, validation, error reporting and CSV export.	**Must**
FR-INV-08	Support physical-stock counts saved as drafts and posted as audited adjustments.	**Should**
FR-INV-09	Prevent duplicate active barcodes and SKUs through application and database validation.	**Must**

5.4 Suppliers and purchases

ID	Requirement	Priority
FR-PUR-01	Create, edit, deactivate and search suppliers with contact, address and GST details.	**Must**
FR-PUR-02	Record supplier invoices with products, quantities, costs, tax, discount, payment status and supplier invoice reference.	**Must**
FR-PUR-03	Increase stock and write stock-ledger entries atomically when a purchase is confirmed.	**Must**
FR-PUR-04	Support paid, partially paid and unpaid purchase states.	**Should**
FR-PUR-05	Prevent duplicate supplier invoice references for the same supplier unless an authorized override is recorded.	**Must**
FR-PUR-06	Support purchase returns that deduct stock, update supplier balance where applicable and require a reason.	**Must**
FR-PUR-07	Attach local invoice images or PDF files and keep them within managed application storage.	**Could**

5.5 Point of sale and billing

ID	Requirement	Priority
FR-POS-01	Use a 70/30 desktop layout: cart on the left and barcode search, totals and checkout on the right.	**Must**
FR-POS-02	Add a product when scanner input ends with Enter; repeated scans increment quantity and unknown barcodes show a recoverable warning.	**Must**
FR-POS-03	Restore focus to barcode entry after item addition and after closing cart or payment dialogs.	**Must**
FR-POS-04	Support quantity increment, decrement, removal, decimal quantity, cart clearing and optional invoice notes.	**Must**
FR-POS-05	Support line-level and bill-level percentage or fixed discounts subject to role limits.	**Must**
FR-POS-06	Calculate subtotal, discount, taxable amount, CGST/SGST or IGST, rounding and final payable amount on the backend.	**Must**
FR-POS-07	Record cash, UPI, card and Udhaar payments; capture tendered cash, change and optional payment reference.	**Must**
FR-POS-08	Support split payment across permitted payment modes.	**Should**
FR-POS-09	At checkout, revalidate product status, price and stock; create invoice, items, payment, stock and Khata updates in one transaction.	**Must**
FR-POS-10	Use an idempotency key so repeated submission returns the original result rather than creating another invoice.	**Must**
FR-POS-11	Support holding and resuming a cart without reserving or deducting stock; revalidate it on resume.	**Should**

ID	Requirement	Priority
FR-POS-12	Persist an optional recoverable cart draft after an abnormal application shutdown without treating it as a sale.	**Must**
FR-POS-13	Implement F1 barcode focus, F2 payment, F4 checkout/print and Esc cart-clear request, plus accessible cart-navigation shortcuts.	**Must**

5.6 Cancellation, returns and refunds

ID	Requirement	Priority
FR-RET-01	Completed invoices shall not be physically deleted.	**Must**
FR-RET-02	An authorized user may cancel an invoice with a mandatory reason, restoring stock and reversing payment and Khata effects.	**Must**
FR-RET-03	Search an original sale by invoice number, customer, phone or date and return selected quantities.	**Must**
FR-RET-04	Prevent cumulative returned quantity from exceeding net sold quantity.	**Must**
FR-RET-05	Reverse applicable discounts and taxes and record the refund mode.	**Must**
FR-RET-06	Return goods to saleable stock or classify them as damaged, with an auditable reason.	**Must**
FR-RET-07	Generate a return receipt or credit note linked to the original invoice.	**Must**

5.7 Customers and Khata

ID	Requirement	Priority
FR-KHA-01	Create, edit, deactivate and search customers by name and phone, with optional address, GSTIN, notes and credit limit.	**Must**
FR-KHA-02	Require a customer for Udhaar and display current due and projected balance before confirmation.	**Must**
FR-KHA-03	Create immutable ledger entries for opening balance, credit sale, settlement, return, cancellation and authorized adjustment.	**Must**
FR-KHA-04	Accept full or partial settlement using cash, UPI, card or other configured mode and generate a receipt.	**Must**
FR-KHA-05	Prevent settlement greater than due unless advance balances are explicitly enabled.	**Must**
FR-KHA-06	Generate a date-filtered customer statement showing invoices, settlements, adjustments and running balance.	**Must**
FR-KHA-07	Provide printable and CSV/PDF statement exports without internet access.	**Should**

5.8 Receipt printing

ID	Requirement	Priority
FR-PRN-01	Discover operating-system printers, configure a default printer and provide a non-financial test print.	**Must**
FR-PRN-02	Support clean monochrome 58 mm and 80 mm thermal receipt profiles.	**Must**
FR-PRN-03	Print shop, invoice, cashier, customer, item, tax, payment and Khata details as applicable.	**Must**
FR-PRN-04	Trigger printing only after the sale transaction commits.	**Must**
FR-PRN-05	Queue failed print attempts locally and allow retry without creating another invoice.	**Must**

ID	Requirement	Priority
FR-PRN-06	Mark reprints as duplicate where configured while retaining the original invoice number and date.	**Must**
FR-PRN-07	Provide PDF or HTML receipt export when a printer is unavailable.	**Should**

5.9 Dashboard and reports

ID	Requirement	Priority
FR-REP-01	Display today's gross sales, net sales, bill count, average bill, payment totals, Udhaar, collections and estimated gross profit.	**Must**
FR-REP-02	Display low-stock, out-of-stock and top-selling product summaries.	**Must**
FR-REP-03	Provide invoice, item, category, payment, GST, discount, cancellation and return reports.	**Must**
FR-REP-04	Provide current stock valuation, stock movement, adjustment, damaged-stock and slow/fast-moving reports.	**Should**
FR-REP-05	Calculate gross profit using purchase cost captured on each sale item, not the product's current cost.	**Must**
FR-REP-06	Support date, cashier, product, category, customer and payment-mode filters.	**Must**
FR-REP-07	Export reports to local CSV and provide print-friendly views; long exports shall show progress and permit cancellation.	**Must**

5.10 Backup, restore, updates and diagnostics

ID	Requirement	Priority
FR-BCK-01	Create scheduled encrypted backups to a configured local, external-drive or network-folder destination.	**Must**
FR-BCK-02	Verify backup creation, retain configurable generations and keep backup failures visible until resolved or acknowledged.	**Must**
FR-BCK-03	Provide "Back up now" with destination selection, progress, result, timestamp and file size.	**Must**
FR-BCK-04	Validate backup integrity, version and password before administrator-authorized restoration.	**Must**
FR-BCK-05	Back up the current database before restore when possible, block normal use during restore and audit the result.	**Must**
FR-UPD-01	Accept signed offline update packages, show current versions and verify package signature and compatibility.	**Must**
FR-UPD-02	Create a pre-update backup, apply Flyway migrations and abort safely when migration validation fails.	**Must**
FR-DIA-01	Show application, database, disk, printer and backup health in a local diagnostics screen.	**Must**
FR-DIA-02	Create a sanitized support bundle containing versions, configuration summary, migration status and rotating logs.	**Must**

6 Business Rules and Data Integrity

ID	Rule
BR-001	The backend is authoritative for price, discount, tax, rounding, stock availability and final payable amount.
BR-002	Money and measurable quantities use decimal arithmetic with centrally configured scale and rounding; binary floating-point types are prohibited.
BR-003	Checkout saves invoice, items, payment, stock transactions and Khata effects within one database transaction.
BR-004	Stock is revalidated under a lock or atomic conditional update immediately before deduction.
BR-005	Products are locked in a consistent order during checkout to reduce deadlock risk.
BR-006	Every stock change requires a stock-ledger entry; direct untracked stock edits are prohibited.
BR-007	A completed financial record cannot be physically deleted or overwritten; corrections use a linked reversal record.
BR-008	Historical sale items retain original selling price, purchase cost, discount and tax values.
BR-009	Invoice numbers are unique and generated by a database-controlled sequence or equivalent concurrency-safe mechanism.
BR-010	Idempotency keys are required for checkout, cancellation, return and Khata settlement.
BR-011	Udhaar requires an identified customer and credit-limit validation according to configuration.
BR-012	A return quantity cannot exceed the remaining returnable quantity from the original sale.
BR-013	Printing and export occur after commit and cannot change the outcome of a financial transaction.
BR-014	All timestamps use the configured shop timezone for display; the system records a stable transaction timestamp and business date.
BR-015	Clock rollback, administrator overrides, backup restore and update operations generate audit events.

6.1 Standard billing calculation order

1. Calculate line base amount: quantity × unit price.
2. Apply authorized line discount.
3. Allocate applicable bill-level discount.
4. Determine taxable amount.
5. Calculate CGST and SGST, or IGST, according to configuration.
6. Aggregate invoice subtotal, discount and tax.
7. Apply configured final rounding.
8. Validate payment allocation equals final payable amount.

> The exact inclusive-versus-exclusive GST policy and rounding scale must be approved before database and API implementation. Test vectors should be captured as executable unit tests.

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Non-Functional Requirements

7.1 Offline operation

ID	Requirement	Acceptance target
NFR-OFF-01	All core operations work with networking disabled.	Complete purchase, sale, return, Khata, print, report and backup acceptance tests offline.
NFR-OFF-02	All runtime assets are installed locally.	No CDN fonts, scripts, styles, icons or mandatory remote APIs.
NFR-OFF-03	Application startup never waits for the internet.	Reach login/POS normally with all network adapters disabled.
NFR-OFF-04	Business data stays local by default.	No outbound business-data transfer unless an optional integration is explicitly configured.

7.2 Performance and capacity

ID	Operation or capacity	Acceptance target
NFR-PER-01	Cold application startup	10 seconds or less on the recommended workstation.
NFR-PER-02	Warm application startup	5 seconds or less.
NFR-PER-03	Barcode lookup	95% complete within 150 ms.
NFR-PER-04	Product search	95% complete within 300 ms for 100,000 products.
NFR-PER-05	Cart recalculation	Visible UI result within 100 ms.
NFR-PER-06	Checkout	95% commit within 1 second, excluding printing.
NFR-PER-07	Current-day dashboard	Load within 2 seconds.
NFR-PER-08	Standard 30-day report	Load within 3 seconds.
NFR-CAP-01	Supported data volume	100,000 products, 1 million invoices and 10 million invoice items without redesign.
NFR-CAP-02	Historical retention	At least five years of ordinary shop data, subject to available disk capacity.

7.3 Reliability and recovery

ID	Requirement	Acceptance target
NFR-REL-01	No partial financial transaction.	Fault-injection tests prove complete commit or rollback.
NFR-REL-02	No duplicate business transaction.	Repeated requests with one idempotency key produce one transaction.
NFR-REL-03	Application crash recovery.	No loss of committed transactions; usable restart within 5 minutes.
NFR-REL-04	Operating-system crash recovery.	Database performs automatic recovery; usable restart within 15 minutes subject to disk integrity.
NFR-REL-05	Printer independence.	Printer cancellation or failure leaves the committed invoice intact and reprintable.
NFR-REL-06	Low-disk protection.	Warn before critical threshold and prevent unsafe exports/backups without corrupting sales.

7.4 Backup and disaster recovery

ID	Requirement	Acceptance target
NFR-BCK-01	Crash recovery point	Zero loss of transactions already committed to healthy storage.
NFR-BCK-02	Device-loss recovery point	No more than 24 hours when daily external backups are maintained.
NFR-BCK-03	Device-loss recovery time	Restore on replacement workstation within 4 hours.
NFR-BCK-04	Backup confidentiality	Every portable backup is encrypted; credentials are excluded from logs.
NFR-BCK-05	Restore verification	Restore tests are performed at least quarterly and before major releases.
NFR-BCK-06	Backup identity	File metadata contains shop ID, time, application version and schema version.

7.5 Security

ID	Requirement	Acceptance target
NFR-SEC-01	Local service isolation	Spring Boot and MySQL bind to loopback only in single-workstation mode.
NFR-SEC-02	Unique installation secrets	No shipped default database password or signing secret.
NFR-SEC-03	Credential protection	Secrets are held in OS-protected storage or a restricted configuration location.
NFR-SEC-04	Desktop process isolation	React renderer has no unrestricted Node.js or filesystem access.
NFR-SEC-05	Authorization	Every protected backend action has role/permission integration tests.
NFR-SEC-06	Password storage	Use an adaptive password hash with per-password salts.
NFR-SEC-07	Update integrity	Unsigned, tampered or incompatible update packages are rejected.
NFR-SEC-08	Sensitive logging	Passwords, tokens, database credentials and unnecessary customer details never appear in logs.
NFR-SEC-09	Auditability	Login, override, adjustment, cancellation, return, restore and export events record user and timestamp.

7.6 Usability and accessibility

ID	Requirement	Acceptance target
NFR-USA-01	Keyboard-first sale	Trained cashier completes a normal sale without a mouse.
NFR-USA-02	Scanner focus	Focus returns correctly after item addition and dialog close.
NFR-USA-03	Desktop fit	Core screens work at 1366×768 without page-level horizontal scrolling.
NFR-USA-04	Visible status	Success, warning and failure states use both color and text/icon meaning.
NFR-USA-05	Error recovery	Recoverable failures preserve the cart and explain the next action.
NFR-USA-06	Destructive actions	Cart clear, cancellation, restore and data removal require clear confirmation.
NFR-USA-07	Focus and contrast	Interactive controls provide visible focus and readable high-contrast presentation.

7.7 Printing and hardware

ID	Requirement	Acceptance target
NFR-PRN-01	Receipt preparation	Complete within 500 ms after checkout response.
NFR-PRN-02	Paper profiles	Verified test output for both 58 mm and 80 mm widths.
NFR-PRN-03	Monochrome readability	Essential content remains legible without color or background graphics.
NFR-PRN-04	Scanner throughput	Rapid sequential scans do not drop or merge barcode events.
NFR-PRN-05	Hardware isolation	Failure of optional hardware adapters cannot stop core billing.

7.8 Maintainability, testing and support

ID	Requirement	Acceptance target
NFR-MNT-01	Modular boundaries	No direct cross-module repository access or circular module dependency.
NFR-MNT-02	Schema control	Every database change is represented by a versioned Flyway migration.
NFR-MNT-03	Critical test coverage	At least 80% branch coverage for sales, stock, payment, return and Khata service logic.
NFR-MNT-04	MySQL verification	Repository, migration and concurrency tests run against MySQL.
NFR-MNT-05	Diagnostics	Support bundle includes sanitized logs, versions, health and migration status.
NFR-MNT-06	Log retention	Structured local logs rotate by size and age and cannot exhaust the disk.
NFR-MNT-07	API consistency	Versioned `/api/v1` endpoints use DTOs and a consistent error contract.
NFR-MNT-08	Architecture records	Material decisions are documented as Architecture Decision Records.

7.9 Packaging and portability

ID	Requirement	Acceptance target
NFR-PKG-01	Clean-machine installation	Install and run without development tools or manual command-line steps.
NFR-PKG-02	Application/data separation	Mutable data is stored outside the read-only program installation directory.
NFR-PKG-03	Upgrade safety	Existing business data is preserved and a pre-update backup is generated.
NFR-PKG-04	Installation size	Target below approximately 1.5 GB excluding business data and backups.
NFR-PKG-05	Recommended workstation	64-bit supported Windows environment, four-core CPU, 8 GB RAM, SSD and at least 5 GB initially free.

8 Data and Integration Requirements

8.1 Core domain records

Area	Principal records	Important relationships
Store and security	Shop, User, Role, Permission, Refresh Session, Audit Event	User receives roles; events reference acting user.
Catalog	Product, Product Barcode, Category, Unit, Tax Rate	Product belongs to category/unit and carries tax configuration.
Inventory	Stock Balance, Stock Transaction, Stock Count	Every stock transaction references product, type and source record.
Purchasing	Supplier, Purchase, Purchase Item, Purchase Payment, Purchase Return	Confirmed purchase items increase stock.
Sales	Invoice, Invoice Item, Payment, Sale Return, Refund	Invoice items snapshot product, price, cost, tax and discount.
Khata	Customer, Khata Entry, Settlement	Credit sales and reversals create linked ledger entries.
Operations	Held Cart, Print Job, Backup Record, Application Setting	Operational records never redefine committed financial history.

8.2 External interfaces

Interface	MVP method	Failure behavior
Barcode scanner	Keyboard-emulation input ending with Enter	Manual barcode entry remains available.
Thermal printer	Operating-system print subsystem	Queue retry; sale remains committed.
CSV files	Local file picker through secure desktop bridge	Preview errors; never partially post invalid import.
Backup storage	Selected folder, removable drive or network folder	Persistent warning; normal billing continues.
Offline updater	Signed installation package	Reject and retain current version on validation failure.
UPI/card terminal	Cashier-confirmed external payment	No automatic online verification in MVP.

8.3 API conventions

- Version endpoints under `/api/v1`.
- Use request and response DTOs; never expose JPA entities.
- Validate all input and return field-level errors.
- Use pagination and bounded filters for large collections.
- Return a consistent error containing code, message, field errors, timestamp, path and correlation ID.
- Publish an OpenAPI description for development and support.

9

Acceptance Scenarios

9.1 Normal cash sale

1. Cashier signs in with the workstation offline.
2. Cashier scans products; repeated scans increment quantity.
3. POS displays current subtotal, discount, GST and payable amount.
4. Cashier presses F2, enters cash received and confirms.
5. System revalidates stock and atomically commits invoice, payment and stock ledger.
6. Receipt prints; barcode focus returns for the next sale.

Pass condition: invoice and stock are correct, no mouse is required, and the flow works with networking disabled.

9.2 Concurrent or duplicate checkout protection

1. Checkout is submitted twice using the same idempotency key.
2. The backend completes or locates the first committed result.
3. Both responses refer to the same invoice.

Pass condition: one invoice, one payment and one stock deduction exist.

9.3 Insufficient stock

1. A cart contains a quantity no longer available at checkout.
2. The backend rejects checkout before creating any financial record.
3. The POS preserves the cart and highlights affected items.

Pass condition: no partial invoice, payment, Khata or stock mutation occurs.

9.4 Udhaar sale and settlement

1. Cashier selects a customer and confirms an Udhaar payment.
2. System validates the credit policy and commits invoice, stock and Khata debit.
3. Later, cashier records a partial cash settlement.
4. System creates an immutable Khata credit and prints a settlement receipt.

Pass condition: running balance and statement match the invoice and settlement.

9.5 Printer failure

1. A valid checkout commits while the printer is unavailable.
2. POS shows “Sale completed; print failed” and retains a retry action.
3. Cashier reconnects the printer and reprints the original invoice.

Pass condition: no duplicate sale is generated and reprint marking follows configuration.

9.6 Crash during checkout

1. The application is forcibly terminated during the checkout request.
2. On restart, the backend checks the idempotency key and transaction state.
3. The POS displays the committed invoice or restores the unsubmitted cart.

Pass condition: the outcome is unambiguous and contains no partial transaction.

9.7 Backup and restore

1. Administrator creates an encrypted backup to a removable drive.
2. The application verifies and records the backup.
3. On a test installation, administrator selects the backup and supplies credentials.
4. The system restores, applies compatible migrations and validates core record counts.

Pass condition: restored invoices, stock and Khata balances reconcile.

9.8 Offline update

1. Administrator selects a signed compatible update package.
2. The system verifies signature, checks disk space and creates a backup.
3. The installer updates application components and applies migrations.
4. The application restarts and reports updated versions.

Pass condition: existing data remains intact; an invalid package is rejected without changing the installation.

Delivery Scope and Roadmap

Stage	Deliverables	Exit condition
1. Foundation	Architecture decisions, domain model, MySQL schema, Flyway, application skeleton, error contract and audit base.	Schema and module boundaries approved; clean startup succeeds.
2. Store & security	First-run setup, local admin, roles, permissions, authentication and protected local API.	Offline login and authorization tests pass.
3. Inventory & purchases	Catalog, units, barcodes, stock ledger, alerts, suppliers, purchases and returns.	Purchase-to-stock and adjustment reconciliation pass.
4. POS & billing	Scanner-first cart, calculation engine, payments, checkout locking, invoice and thermal receipt.	Cash-sale, duplicate and insufficient-stock scenarios pass.
5. Khata & returns	Customers, credit sales, settlements, statements, cancellation, sales return and refund.	Ledger and reversal reconciliation pass.
6. Reports	Dashboard, sales, profit, tax, stock and export reports.	Reports reconcile with test transactions.
7. Desktop operations	Installer, managed services, printing, encrypted backup/restore, updater and diagnostics.	Clean-machine, crash, restore and update tests pass.
8. Release hardening	Performance, security, concurrency, packaging and user-acceptance testing.	All Must requirements and critical NFR targets pass.

Future roadmap

Phase 2 candidates

- Cashier shift reconciliation
- Expense recording
- Advanced barcode label printing
- Batch and expiry tracking
- Direct silent-print adapters
- Optional weighing-scale adapter

Future platform capabilities

- Local-network multi-terminal mode
- Cloud synchronization and remote reports
- Multi-store inventory
- Customer messaging integrations
- E-invoice integrations
- Mobile companion application

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Risks and Open Decisions

11.1 Principal risks

Risk	Impact	Mitigation
Bundled desktop footprint	Electron, Java and MySQL produce a relatively large installer.	Use a minimized Java runtime, exclude development artifacts and set an explicit size budget.
Database lifecycle complexity	Failed MySQL startup can block the application.	Managed startup, health checks, recovery UI, safe logs and tested repair procedures.
Single-device failure	Disk or computer loss can stop business and lose unbacked-up data.	Visible external-backup status, encrypted daily backups and documented replacement-device restore.
Thermal printer diversity	Margins and silent-print behavior vary by driver.	OS printing for MVP, 58/80 mm profiles, test-print wizard and limited certified-printer list.
Incorrect computer time	Financial reports and invoice order may be misleading.	Clock-rollback detection, restricted backdating and audit trail.
Tax-rule ambiguity	Incorrect GST or rounding affects bills and reports.	Approve calculation policy and test vectors before implementation.
MySQL redistribution licensing	Bundling MySQL Community server binaries may impose distribution obligations.	Complete project-specific licensing review before packaging MySQL; obtain an appropriate license or revise the installation model.
Premature multi-terminal scope	Synchronization and conflict handling delay the MVP.	Keep one authoritative database; add LAN clients only as a separate architecture phase.

11.2 Decisions requiring approval

ID	Decision	Recommended baseline
DEC-01	Supported operating-system editions	One documented 64-bit Windows target for the first release.
DEC-02	MySQL lifecycle	Application-managed local service/instance bound to loopback.
DEC-03	Printing method	OS print flow first; certified silent printing later.
DEC-04	Negative stock	Disallow by default; optional audited administrator override only if required.
DEC-05	GST price mode	Choose inclusive or exclusive defaults and approve rounding examples.
DEC-06	Invoice numbering	Financial-year prefix plus monotonic local database sequence.
DEC-07	Backup encryption and owner recovery	Owner-defined backup password with documented recovery responsibility.
DEC-08	LAN mode	Exclude from MVP; design as a later centrally hosted shop-server profile.
DEC-09	MySQL distribution model	Complete licensing review before bundling server binaries in the desktop installer.

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Glossary and Approval

Term	Meaning
Atomic transaction	A group of database changes that all commit together or all roll back.
Barcode scanner	A device that normally sends scanned text as keyboard input.
Business date	The shop-accounting date assigned to a transaction.
CGST / SGST / IGST	Configured GST components recorded on taxable transactions.
Idempotency key	A unique request key that makes a retried financial operation return the same result.
Khata	The customer-credit ledger for Udhaar and subsequent settlements.
Loopback	Network access restricted to the same computer, normally through `127.0.0.1`.
Modular monolith	One deployable backend divided into strongly separated business modules.
RPO	Recovery Point Objective: maximum tolerable data loss measured in time.
RTO	Recovery Time Objective: target time to restore service.
Stock ledger	Immutable history explaining every increase or decrease in stock.
Udhaar	A sale amount recorded as credit owed by an identified customer.

> **Recommended next artifact:** approve the open decisions, then create the conceptual data model, table constraints and Flyway migration plan before scaffolding the application.

Approval record

Product owner / shop representative — name, signature and date

Solution architect / engineering lead — name, signature and date

End of Software Requirements Specification — Version 1.0